WHAT QUESTIONS THE MVP SOLVES AND PROVES.

Cryptographic Protocol of Sovereign Collective Decision-Making.

1. Existing Approach and Its Limitations.

Current manipulation-detection systems in decentralised environments fall into four classes:

Class	Representatives	Object of Analysis
Static code analysis	Slither, MythX, OpenZeppelin	Smart contract code
Rules and signatures	Forta Network	On-chain events in real time
Commercial intelligence	Chainalysis, Elliptic, TRM Labs	Transaction graph, addresses
Dynamic time-series analysis	Huynh et al. (arXiv:2308.16391)	Address behaviour over time

- *Fundamental limitation of all four classes: manipulation is detected after it has already occurred. Defence is reactive. The object of analysis is the external trace of actions — not the judgment itself.*

BeTrueCore offers an orthogonal solution: complete exclusion of conditions for manipulation through immutable mathematical predicates — before the vote is cast.

2. Comparison of Approaches.

Criterion	Existing Approach	BeTrueCore
Philosophy of protection	Reactive: detection after the fact	Architectural: exclusion of conditions for attack
Object of analysis	Transactions, flows, frequency	Behavioural trace: timings, VWU history, participation depth
Intermediate results	Available in real time	Absent — informational isolation of the participant
Temporal factor	Time-series of contract lifecycle	Dynamic VWU trajectory + missed opportunity notification
Defence engine	Classical ML classifiers	Sentinel + Strategist + Analyst

3. Basis of BeTrueCore.

3.1 Conceptual Shift.

The mathematical and philosophical basis of the system — transferring privacy from the plane of "policy" (trust in institutions) to the plane of "predicate" (a mathematical property verifiable without identity disclosure).

3.2 Architecture L0–L5.

Layer	Component	Function
L0 Identity	State identifier (NIN-code) + on-device ZK transformation + MPC 2-of-3 key splitting	Uniqueness without disclosure. NIN-code never leaves the participant's device.
L1 Proofs	zk-SNARK, Circom/Groth16 (MACI anti-collusion pattern), ZK-nullifier	Anti-collusion. Receipt-freeness.
L2 Execution	VWU contract, Optimism L2	Judgment weight calculation and update.
L3 Lock	Lit Protocol, time-lock	Synchronous disclosure — Panorama Effect at 20:00.
L4 Archive	Celestia DA, Kintsugi	Immutable archive independent of domain.
L5 AI Agents	9 agents (3 Analyst + 3 Strategist + 3 Sentinel), read-only	Observation with no right to modify state. AI as notary, not judge.

3.3 Operator and Trials of Decentralisation.

The Coordinator is a technical node that decrypts votes, tallies the result, and publishes a ZK proof of correctness to the blockchain. It sees the votes but cannot falsify them — a fraudulent proof is automatically rejected by the protocol.

In Trial 1, the human hand is needed exactly once: to launch the system and hand control over to the mathematics. After that, no further human involvement in execution is required.

Trial	Operator	Nature of Trust
Trial 1 (MVP)	BeTrueCore — founder / team	Human executes with mathematical support.
Trial 2	Lit Protocol MPC quorum (k-of-N)	Mathematics becomes the operator. No single person can unilaterally control

		the outcome.
Trial 3	Full decentralisation	Community governs through the protocol itself.

Three levels of result protection:

- ✓ A single node — cannot change anything
- ✓ A quorum k-of-N — can block availability, but cannot falsify the result
- ✓ Falsification of the result — mathematically impossible: the ZK proof is verified by the protocol automatically, regardless of the number of nodes

❖ *Analogy — Odysseus at the mast. Trial 2 is the moment BeTrueCore ties itself to the mast. After that, the mathematics executes the founder's decision — even if he later wishes to change it.*

3.4 Three Critical Vulnerabilities and Architectural Responses.

Subjectivity at launch (oracle problem):

The 23×32 matrix (736 cells) is chosen by a human. This is a documented "stake" — not a hidden flaw but an explicit architectural assumption recorded in Zenodo.

Thermostat vulnerability:

Thermostat vulnerability: The VWU weight update channel is monitored by Sentinel: upon detection of anomalous coordination, the anomaly is flagged to Celestia DA as a public audit record. VWU updates are not blocked — the system informs, it does not penalise.

Byzantine Fault at the MACI Coordinator:

A compromised coordinator affects only availability, not data integrity. L5 operates strictly in read-only mode.

3.5 Principle of Architectural Resilience.

The system creates an architecturally resilient and economically unfeasible-to-attack combination of four independent factors:

- ✓ Timestamps — every step is recorded at a precise moment. A bot can imitate an action but cannot predict exactly when to act.
- ✓ Behavioural history (VWU) — each participant's weight is their personal trajectory built over months of real participation. It cannot be synthetically reproduced.
- ✓ No intermediate results — the participant makes decisions in informational isolation. There is no signal for an algorithm to calibrate against.
- ✓ Collective TOP-3 — no participant knows in advance which dilemmas will appear in the Panorama.

❖ *The combined cost of simultaneously breaching all independent protection layers exceeds any reasonable stake in shifting an outcome.*

3.6 Architectural Indivisibility.

VWU without ZK-anonymity, without MACI receipt-freeness, without the absence of intermediate results, and without the 736-point matrix is simply an activity counter. Thousands of those already exist. Zero value.

The system works only as a whole — or not at all. Partial copying is impossible by definition: without each layer, the others lose meaning. This is defensibility — protection from replication without a patent and without a secret.

4. AI Sentinel — Operating Principle.

4.1 The Seven-Step BeTrueCore Cycle.

Each session consists of two phases:

Step	Phase	Action
1	Prompt	Participant creates a prompt — a dilemma for the community
2–4	Prompt	Voting for prompts proposed by the community
5–7	Dilemma	Binary choice on dilemmas from the community-selected TOP-3 Panorama

- *The TOP-3 is formed collectively by the community. The participant's own prompt may or may not enter the TOP-3. A participant may stop at any step — from 0 to 7.*
  - Code implementation — Appendix 1 (BeTrueCoreL5Engine).

4.2 How Sentinel Detects Attacks.

Sentinel analyses not individual participants but the degree of correlation between sessions over time. Honest users act chaotically and unpredictably. A coordinated attack with high probability creates a hidden mathematical correlation.

Two key signals:

- ✓ Spectral entropy — a measure of independence of participant actions. During a coordinated attack it typically drops sharply. The Harmony Agent switches the pool status to RED or SUPERPOSITION.
- ✓ Participation depth — an additional behavioural signal against the current generation of attacks. Scripts as a rule complete the full cycle. A real person is unpredictable: 2/7 today, 7/7 tomorrow. The system evolves alongside the threats.

  - Code implementation — Appendix 2 (AI Sentinel L5)

4.3 The 736-Rule Matrix and Traffic Light.

736 rules — the intersection of 23 Asilomar Principles and 32 TDSH parameters. Each cell is a formal rule for the agents. The Harmony Agent computes an Ematch score and issues a verdict:

Ematch	Verdict	Meaning
[0.0 — 0.4)	RED	Integrity threshold breached — critical degradation
[0.4 — 0.7)	YELLOW	Partial degradation — attention required
[0.7 — 1.0]	GREEN	Collective judgment integrity confirmed

The verdict is applied equally to the majority and the minority. Both results appear in the Panorama with colour. The matrix does not choose sides — it measures both.

736 + traffic light provide protection at two levels simultaneously:

- ✓ Inside the process — detects anomalous coordination
- ✓ The process itself — measures the quality of collective judgment regardless of community size

Source: [9] The Ethical Priority Map, Zenodo DOI: 10.5281/zenodo.21225420

5. VWU — Judgment Weight.

5.1 Principle.

VWU (Verified Wisdom Unit) is a discrete weight unit accumulated through verified quality of ethical judgments over time. Weight is determined not by tokens but by participation history.

Update upon session completion (Panorama, 20:00):

$$W_i^{(t)} = (1 - \alpha) * W_i^{(t-1)} + \alpha * P(X_i^{(t)} \mid Y^{(t)})$$

- $W_i^{(t)}$ — participant weight at step t
- α — exponential smoothing coefficient
- $P(X_i^{(t)} \mid Y^{(t)})$ — likelihood of participant's judgments relative to the Panorama consensus vector

❖ *The thermostat calibration range is a protected parameter disclosed only to developers under NDA.*

VWU protection is built not through blocking but through five structural architectural properties:

P1 Non-linearity of growth — each subsequent unit requires more genuine participation

P2 Optional factor — inactive days do not change the weight; it is frozen at the level reached

P3 Sensitivity to paradoxical patterns — Sentinel tracks long-term contradictions in a participant's choice history

P4 Impossibility of instantaneous accumulation — VWU is a function of time and quality, not a single action

P5 Unreachability through observation — you can see the weight distribution but cannot copy the trajectory

- *Source: [8] The Notary Under Attack. Zenodo DOI: 10.5281/zenodo.21111544*

5.2 Philosophy of Inactivity.

If a participant does not open the application — their weight does not change and is preserved at the level reached. The system sends a missed opportunity notification. Silence is also a sovereign decision.

Missed sessions are missed growth, not a penalty. The formula updates weight only through real participation.

Optional factor (structural property 5.2):

On returning after a prolonged absence nothing is reduced: both the balance and the trajectory are preserved, and the next session is computed with the full delta.

Missed sessions appear in the participant's own report as potential losses — the growth foregone over the preceding 90 days.

The only mechanism that lowers a weight is the Rational filter. It is triggered by manipulation of the gamification — mutually exclusive or purely chaotic choices within one goal and one time limit — not by absence.

State	Weight	System response
Did not participate (0 steps)	Unchanged. Preserved.	Missed opportunity notification
Partial participation (1–6)	Updated via Bayes. Absence does not reduce the delta.	Verdict GREEN / YELLOW / RED
Full cycle (7 steps)	Updated via Bayes. Absence does not reduce the delta.	Verdict GREEN / YELLOW / RED
Sentinel anomaly	Not blocked. Flag written to Celestia DA.	log_only → public audit

- *The system informs, it does not penalise. Silence is also a sovereign decision.*
  - *Code implementation — Appendix 3 (VWU Bayesian Update Layer.*

6. White Feather Effect.

Phase 2+: activated only upon SUPERPOSITION state.

The White Feather Effect uses Goedel not as a strict mathematical derivation but as an architectural principle: the system accepts its incompleteness as a design constraint — rather than concealing it.

Gödel's theorem shows: there exist true statements that cannot be proven from within the system. BeTrueCore accepts this — "AI as notary, not judge" — as a foundation for design. When the system enters SUPERPOSITION state (an irresolvable ethical paradox or Byzantine failure), instead of halting, a stochastic exit is activated via the Ornstein-Uhlenbeck equation.

A system that knows its own incompleteness is more resilient than one that pretends to be complete.

- *"Odysseus tied himself to the mast before he heard the Sirens. The White Feather Effect is what happens when the rope breaks — and the mast still holds course."*
  - *Code implementation — Appendix 4 (Stochastic Exit Layer — White Feather Effect).*

7. What the MVP Solves and Proves.

7.1 MVP — Three Mandatory Components (Phase 1).

Circom/Groth16 smart contract (MACI anti-collusion pattern) + ZK identity circuit + VWU contract.

Without these three components the system does not function. They are indivisible. Celestia and Lit Protocol are added in Phase 2.

WebSocket subscription to the smart contract event (eth_subscribe) is required from day one of MVP development. Polling the blockchain at 20:00 with 50,000+ users will exhaust RPC rate limits.

7.2 Six Questions Closed by MVP.

#	Question	How it is proved	Layer
1	Unique citizen without identity disclosure	NIN-code → ZK commitment → nullifier; test DoubleRegisterFails.	L0 → L1
2	Vote is non-purchasable	Key-rotation → receipt-freeness; test NullifierPreventsDoubleVote.	L1 → L2
3	Judgment weight works as a smart contract	Non-linear growth through regular participation. Missed sessions — missed growth. Test VWU_AbsenceDoesNotReduce.	L2
4	VWU result differs from simple majority	Manual pilot (three dilemmas — divergence documented).	L2
5	Three primitives	Integration test on testnet.	L0–L2

	compatible in one chain		
6	Collapse at 20:00 does not cause overload	WebSocket subscription (eth_subscribe) — system delivers the Panorama at the moment of readiness, without manual page refresh.	L3

7.3 What Each Development Phase Adds.

Phase	Question	Instrument
Phase 1	Collapse is simultaneous across all layers.	20:00 — baseline hypothesis. Staggered timing (18:00 / 19:00 / 20:00) to be tested in Phase 3 based on engagement data.
Phase 2	Mathematics becomes the operator.	Lit Protocol MPC.
Phase 2	Thermostat protected from drift.	Bayesian update + Sentinel block.
Phase 2	Goedel deadlock has an architectural exit.	Ito equation (Ornstein-Uhlenbeck).
Phase 3	Sybil attack detected mathematically.	Spectral entropy SVD (Sentinel L5).
Phase 3	Participation depth — behavioural signal.	participation_depth in session vector.
Phase 3	Unpredictability proven empirically.	Pilot of 50–100 participants.

7.4 Argument for Small and False Communities.

The MVP does not require a guarantee of community purity. It requires only one thing — that the matrix measures correctly.

A false community exposes itself:

Fictitious accounts do not know in advance what will enter the TOP-3. Without that knowledge it is impossible to form a judgment that does not degrade along TDSH parameters. Ematch drops → RED. This is not a vulnerability of the MVP — it is a demonstration of its correct operation.

The honest minority remains visible:

The verdict is applied separately to the majority and the minority. The Panorama shows both colours side by side. The majority does not suppress the minority — each receives its own quality-of-judgment score.

No one can prove anything to anyone:

Receipt-freeness (MACI): neither a vote buyer can verify what they purchased, nor the organiser of a fictitious community can verify that their participants are following instructions. A conspiracy without a guarantee is not a conspiracy.

The exploit destroys itself:

To imitate quality judgment — you must produce quality judgment. To build a false community — you must build a real one. By the time an attacker accumulates enough VWU weight to influence the outcome — they have already become a genuine participant.

- *"To simulate a community, you must practically build one. To successfully simulate a high-weight participant, the adversary must actually become one."*
  [6] ethresear.ch/t/designing-infrastructure-where-exploits-destroy-themselves/25348

- *From [9] Zenodo DOI:10.5281/zenodo.21225420:*
  "The size of the community changes the volume of data — not the structure of measurement."

- *The matrix works identically with 50 participants and with 50,000. A pilot of 50 people is sufficient to prove this.*

7.5 Unified Mathematical Framework.

Three layers of protection form an unbroken chain:

L0/L5 — Sybil attack protection:

736 rules + traffic light protect both inside the process (detection of anomalous coordination) and the process itself (measurement of judgment quality applied equally to majority and minority).

L2/L3 — Weight thermostat protection:

Bayesian VWU update within the protected thermostat corridor. Inactivity is not penalised — weight is preserved. Missed sessions are missed growth. The system informs, it does not penalise.

Stochastic exit — White Feather Effect:

Goedel as an architectural principle, not a strict derivation: the system accepts its incompleteness and exits the deadlock via the Ito equation.

Principle of architectural indivisibility:

The system works only as a whole — or not at all. Partial copying is impossible by definition: without each layer, the others lose meaning. This is defensibility — protection from replication without a patent and without a secret.

MVP proves: sovereign collective decision-making technically exists — not as a concept, but as a verifiable smart contract on testnet. Everything else is built on top of this proof.

- Trial 1: human executes with mathematical support.
- Trial 2: mathematics becomes the operator.
- Trial 3: community governs through the protocol itself.